ISSUE 001 · NOVEMBER 2026

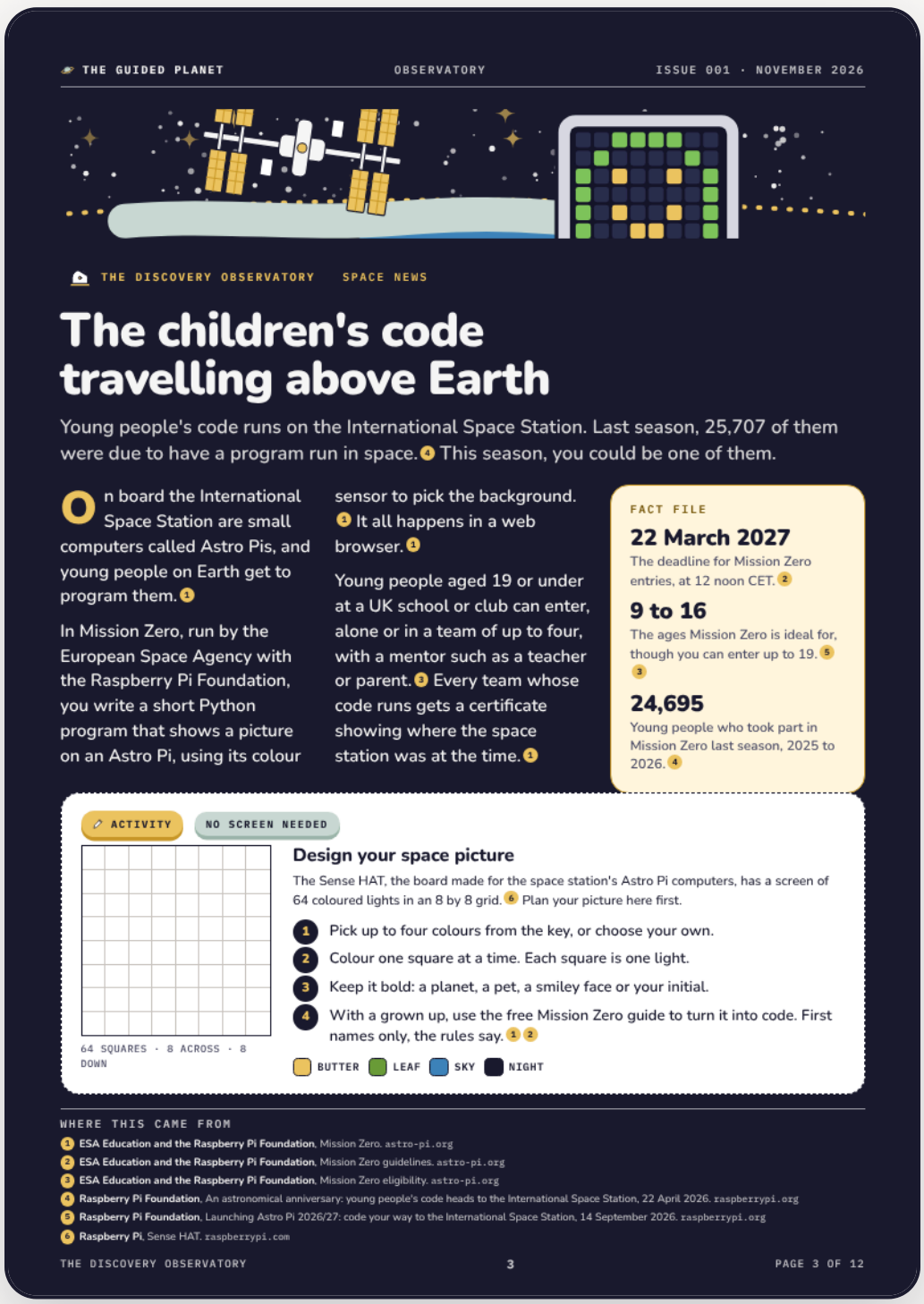
The good side of screens.

A free newspaper for children 7 to 12, and the grown ups who read with them.



The children's code travelling above Earth

Young people's code runs on the International Space Station. Last season, 25,707 of them were due to have a program run in space. This season, you could be one of them.



THE GUIDED PLANET · OBSERVATORY · ISSUE 001 · NOVEMBER 2026

THE DISCOVERY OBSERVATORY · SPACE NEWS

The children's code travelling above Earth

Young people's code runs on the International Space Station. Last season, 25,707 of them were due to have a program run in space. This season, you could be one of them.

On board the International Space Station are small computers called Astro Pis, and young people on Earth get to program them.

In Mission Zero, run by the European Space Agency with the Raspberry Pi Foundation, you write a short Python program that shows a picture on an Astro Pi, using its colour sensor to pick the background. It all happens in a web browser.

Young people aged 19 or under at a UK school or club can enter, alone or in a team of up to four, with a mentor such as a teacher or parent. Every team whose code runs gets a certificate showing where the space station was at the time.

FACT FILE

22 March 2027
The deadline for Mission Zero entries, at 12 noon CET.

9 to 16
The ages Mission Zero is ideal for, though you can enter up to 19.

24,695
Young people who took part in Mission Zero last season, 2025 to 2026.

ACTIVITY · NO SCREEN NEEDED

Design your space picture

The Sense HAT, the board made for the space station's Astro Pi computers, has a screen of 64 coloured lights in an 8 by 8 grid. Plan your picture here first.

64 SQUARES · 8 ACROSS · 8 DOWN

1. Pick up to four colours from the key, or choose your own.
2. Colour one square at a time. Each square is one light.
3. Keep it bold: a planet, a pet, a smiley face or your initial.
4. With a grown up, use the free Mission Zero guide to turn it into code. First names only, the rules say.

BUTTER · LEAF · SKY · NIGHT

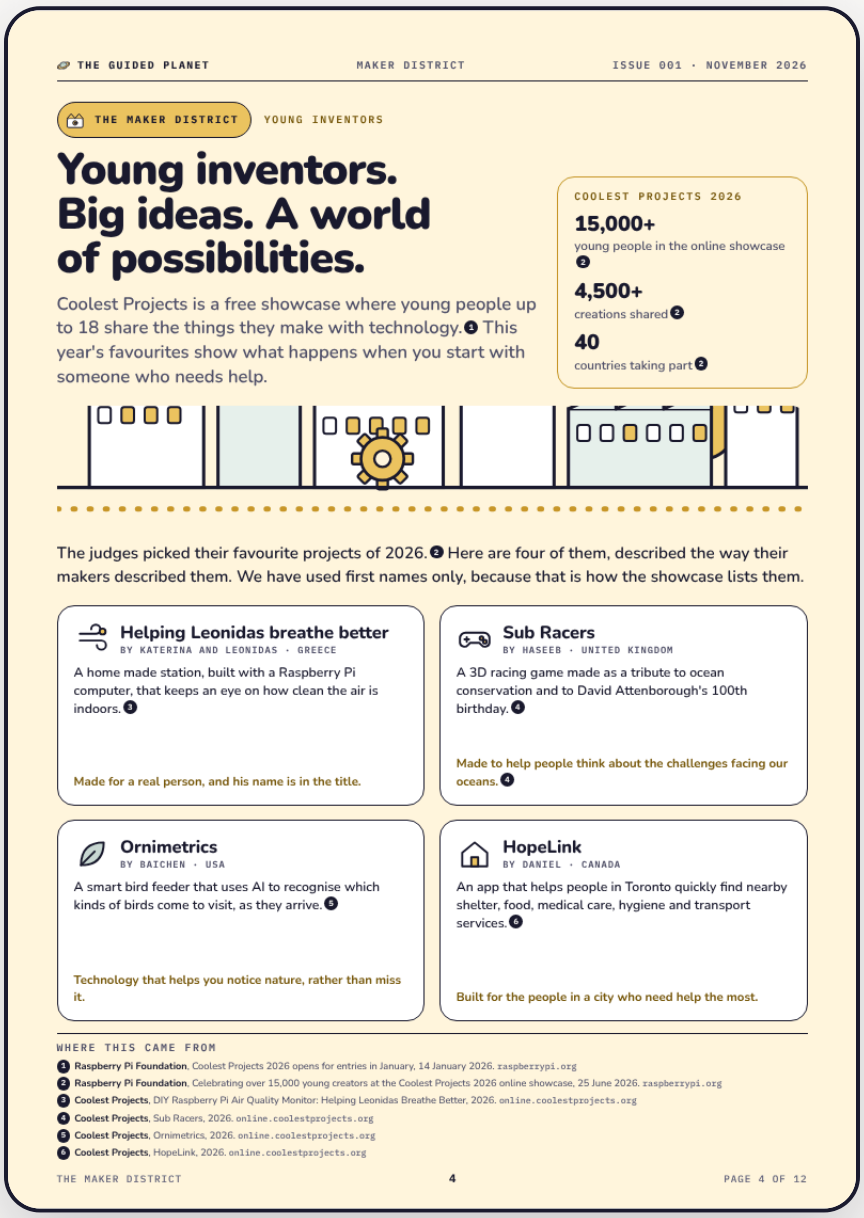
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2. **ESA Education and the Raspberry Pi Foundation**, Mission Zero guidelines. astro-pi.org
3. **ESA Education and the Raspberry Pi Foundation**, Mission Zero eligibility. astro-pi.org
4. **Raspberry Pi Foundation**, An astronomical anniversary: young people's code heads to the International Space Station, 22 April 2026. raspberrypi.org
5. **Raspberry Pi Foundation**, Launching Astro Pi 2026/27: code your way to the International Space Station, 14 September 2026. raspberrypi.org
6. **Raspberry Pi**, Sense HAT. raspberrypi.com

THE DISCOVERY OBSERVATORY · 3 · PAGE 3 OF 12

Young inventors. Big ideas. A world of possibilities.

Coolest Projects is a free showcase where young people up to 18 share the things they make with technology. This year's favourites show what happens when you start with someone who needs help.



THE GUIDED PLANET · MAKER DISTRICT · ISSUE 001 · NOVEMBER 2026

THE MAKER DISTRICT · YOUNG INVENTORS

Young inventors. Big ideas. A world of possibilities.

Coolest Projects is a free showcase where young people up to 18 share the things they make with technology.[1] This year's favourites show what happens when you start with someone who needs help.

COOLEST PROJECTS 2026

15,000+
young people in the online showcase[2]

4,500+
creations shared[2]

40
countries taking part[2]

The judges picked their favourite projects of 2026.[2] Here are four of them, described the way their makers described them. We have used first names only, because that is how the showcase lists them.

Helping Leonidas breathe better
BY KATERINA AND LEONIDAS · GREECE

A home made station, built with a Raspberry Pi computer, that keeps an eye on how clean the air is indoors.[3]

Made for a real person, and his name is in the title.

Sub Racers
BY HASEEB · UNITED KINGDOM

A 3D racing game made as a tribute to ocean conservation and to David Attenborough's 100th birthday.[4]

Made to help people think about the challenges facing our oceans.[4]

Ornimetrics
BY BAICHEN · USA

A smart bird feeder that uses AI to recognise which kinds of birds come to visit, as they arrive.[5]

Technology that helps you notice nature, rather than miss it.

HopeLink
BY DANIEL · CANADA

An app that helps people in Toronto quickly find nearby shelter, food, medical care, hygiene and transport services.[6]

Built for the people in a city who need help the most.

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2. **Raspberry Pi Foundation**, Celebrating over 15,000 young creators at the Coolest Projects 2026 online showcase, 25 June 2026. raspberrypi.org
3. **Coolest Projects**, DIY Raspberry Pi Air Quality Monitor: Helping Leonidas Breathe Better, 2026. online.coolestprojects.org
4. **Coolest Projects**, Sub Racers, 2026. online.coolestprojects.org
5. **Coolest Projects**, Ornimetrics, 2026. online.coolestprojects.org
6. **Coolest Projects**, HopeLink, 2026. online.coolestprojects.org

THE MAKER DISTRICT · 4 · PAGE 4 OF 12

Not all screen time tells the same story

Grown ups often ask how long is too long. Scientists are finding a more useful question: what are you actually doing on the screen, and who are you doing it with?



THE GUIDED PLANET · THE RESEARCH ROOM · ISSUE 001 · NOVEMBER 2026

THE RESEARCH ROOM · WHAT SCIENTISTS KNOW

Not all screen time tells the same story

Grown ups often ask how long is too long. Scientists are finding a more useful question: what are you actually doing on the screen, and who are you doing it with?[1]

Creating
Coding a game, making music, drawing a comic.
You are the one making the choices.

Learning
A how to video, a maths game, a space website.
In one big review, watching together with a parent was linked with better reading and writing skills.[1]

Connecting
A video call with Grandma, a message to a friend.
Talking with people you know is different from scrolling past strangers. See page 7.

Watching
Shows, videos, scrolling through a feed.
Fun, and worth noticing: did you choose it, or was it chosen for you? Page 8.

WHAT THE RESEARCH FOUND

In 2024, scientists brought together 102 earlier research summaries covering almost two million young people.[1] Some kinds of screen use were linked with small to moderate harms and some with benefits, so they advised weighing up each kind separately.[1]

One example: screen time was linked with weaker reading and writing skills overall, but when parents watched together with their children, the link went the other way.[1]

WHAT IT CANNOT TELL US YET

Most of these studies spot links, not causes: two things happening together, not one making the other happen.[2] Many also rely on people estimating their own screen time.[2]

Simple labels are argued about too. A review of 40 studies, mostly of adults, found little support for 'active' social media use being good and 'passive' use being bad.[3] It tells us little about children your age.

TALK ABOUT IT TOGETHER

1. Think about yesterday. Which of the four kinds did you do most?
2. Which kind leaves you feeling good? Which is hard to stop?
3. What is one thing you would like to make, not watch, this week?

DIGI SAYS
Two things happening together does not mean one caused the other. Spotting that is a skill for life.

WHERE THIS CAME FROM

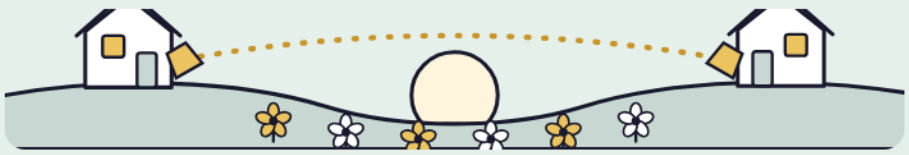
1. **Sanders, Noetel, Parker and colleagues, Nature Human Behaviour**, An umbrella review of the benefits and risks associated with youths' interactions with electronic screens, 2024. doi.org
2. **Kaye, Orben, Ellis, Hunter and Houghton, IJERPH**, The Conceptual and Methodological Mayhem of 'Screen Time', 2020. doi.org
3. **Valkenburg, van Driel and Beyens, New Media and Society**, The associations of active and passive social media use with well-being: A critical scoping review, 2022. doi.org

THE RESEARCH ROOM · 6 · PAGE 6 OF 12

Sometimes a screen helps a friendship travel further

A video call can bring a cousin from across the country into your kitchen. Scientists are learning when screens help friendships, and when they get in the way.

THE GUIDED PLANET · CONNECTION GARDEN · ISSUE 001 · NOVEMBER 2026



THE CONNECTION GARDEN · FRIENDSHIP

Sometimes a screen helps a friendship travel further

A video call can bring a cousin from across the country into your kitchen. Scientists are learning when screens help friendships, and when they get in the way.

Lots of families use video calls and messages to stay close to people who live far away, like grandparents, cousins or a friend who has moved.

In one experiment, 130 young people aged about 12 did a stressful task. The ones who then texted a friend they knew were in a better mood afterwards than the ones who sat quietly or watched a video. [1] It was one short experiment, so it does not prove that more messaging is always better.

A UNICEF project that studied children aged 8 to 12 found that digital games can support feeling connected to other people, and that different games support different things depending on how they are designed. [2]

KEEP YOUR GARDEN SAFE

- Connect with people you know in real life, and let a grown up know who.
- A message from someone you do not know? Tell a grown up. You are never in trouble for telling.
- Many apps, including TikTok, are for ages 13 and over. [3]



PEBBLE SAYS

A kind message can travel a very long way. Who would smile if they heard from you today?

ACTIVITY · NO SCREEN NEEDED

Send a friendship postcard

Cut it out, draw on the front, write on the back, then post it or hand it over.

DRAW SOMETHING YOU DID TOGETHER

DEAR

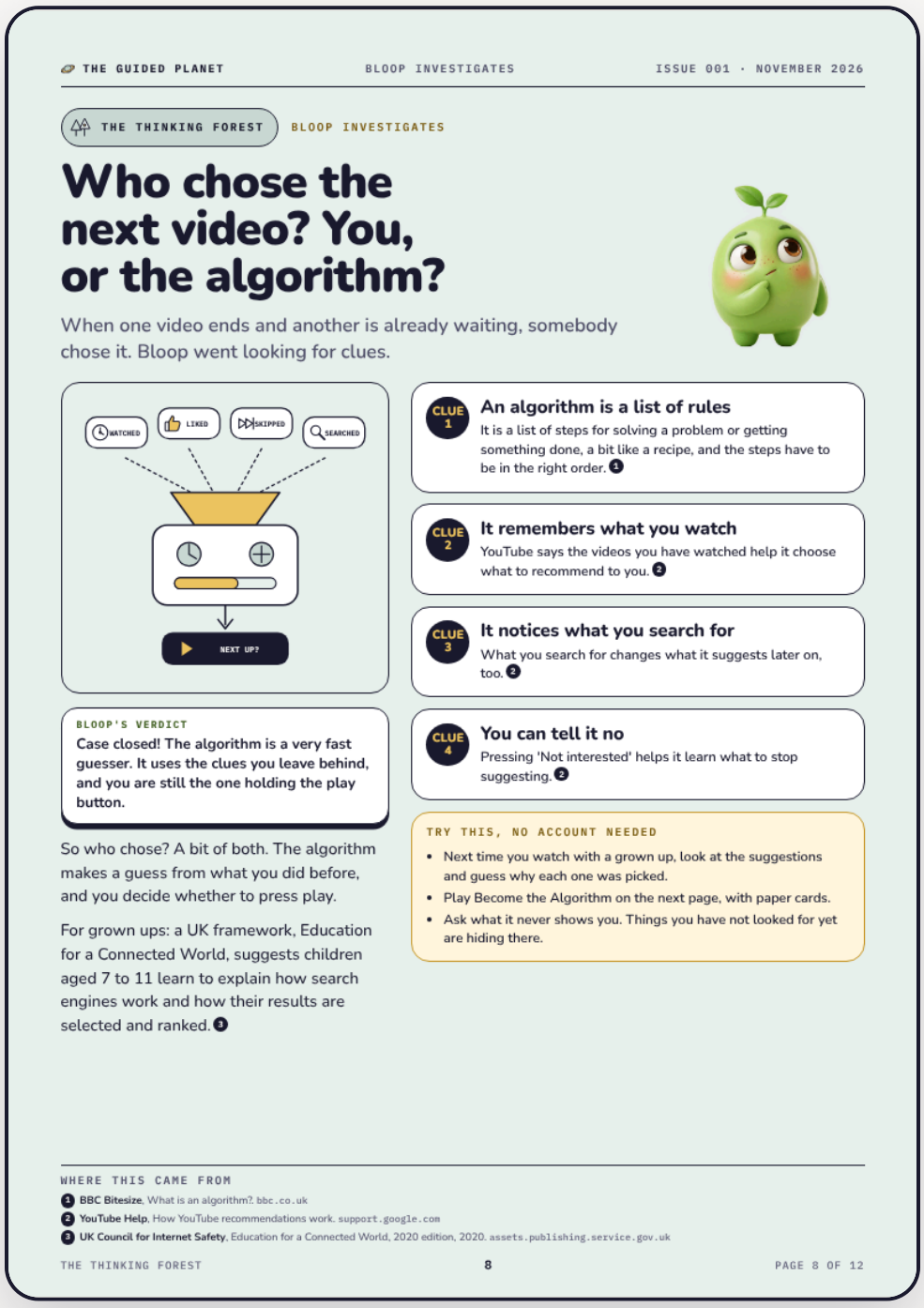
Add a stamp and an address with a grown up · Or take a photo of it and send it together

WHERE THIS CAME FROM

1. **Yau, Reich and Lee, Journal of Adolescent Health**, Coping With Stress Through Texting: An Experimental Study, 2021. doi.org
2. **UNICEF Innocenti**, Responsible Innovation in Technology for Children (RITEC), phase 2 report, April 2024. unicef.org
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Who chose the next video? You, or the algorithm?

When one video ends and another is already waiting, somebody chose it. Bloop went looking for clues.



THE GUIDED PLANET · BLOOP INVESTIGATES · ISSUE 001 · NOVEMBER 2026

THE THINKING FOREST · BLOOP INVESTIGATES

Who chose the next video? You, or the algorithm?

When one video ends and another is already waiting, somebody chose it. Bloop went looking for clues.

CLUE 1 — An algorithm is a list of rules
It is a list of steps for solving a problem or getting something done, a bit like a recipe, and the steps have to be in the right order.[1]

CLUE 2 — It remembers what you watch
YouTube says the videos you have watched help it choose what to recommend to you.[2]

CLUE 3 — It notices what you search for
What you search for changes what it suggests later on, too.[2]

CLUE 4 — You can tell it no
Pressing 'Not interested' helps it learn what to stop suggesting.[2]

BLOOP'S VERDICT

Case closed! The algorithm is a very fast guesser. It uses the clues you leave behind, and you are still the one holding the play button.

So who chose? A bit of both. The algorithm makes a guess from what you did before, and you decide whether to press play.

For grown ups: a UK framework, Education for a Connected World, suggests children aged 7 to 11 learn to explain how search engines work and how their results are selected and ranked.[3]

TRY THIS, NO ACCOUNT NEEDED

- Next time you watch with a grown up, look at the suggestions and guess why each one was picked.
- Play Become the Algorithm on the next page, with paper cards.
- Ask what it never shows you. Things you have not looked for yet are hiding there.

WHERE THIS CAME FROM

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2 **YouTube Help**, How YouTube recommendations work. support.google.com
3 **UK Council for Internet Safety**, Education for a Connected World, 2020 edition, 2020. assets.publishing.service.gov.uk

FREE TO PRINT

Save it for the weekend. Send it to the family whose kids love space.

Twelve pages. No phone or account needed for any activity.